propSym, a new software tool to aid material science research

A new software tool can accelerate materials science research by cutting out tedious background research on material properties. Penn State and Sandia National Laboratories researchers recently debuted propSym, an open source software on the programming platform MATLAB, to calculate the fundamental constants needed to describe the physical properties of solids, such as metals, ceramics, or composites.

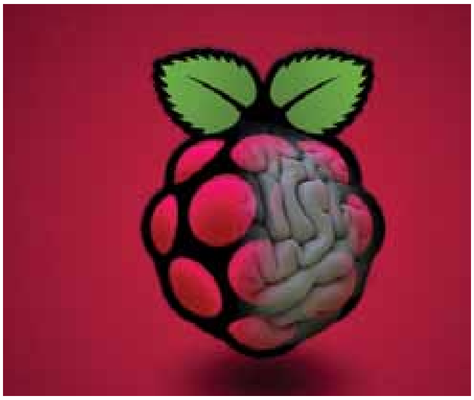
The researchers developed propSym, the details of which were published in the *Journal of Applied Crystallography*, after they could not find reliable information about langasite, a material used in sensing and energy harvesting devices in a separate joint study with Sandia National Labs.

"Traditionally, the relationships between fundamental constants and material symmetries are found only in appendices of textbooks or tables in journal articles," said Christopher Kube, assistant professor of engineering science and mechanics at Penn State, who led the project. "After a thorough search, we were not able to find reference data for several non-linear material properties for langasite. When data was available, we found instances of typos and inconsistencies across references. Incorrect input data will ruin a model."

Kube and his collaborators used propSym to determine the properties of langasite, such as elasticity and the ability to accumulate electric charge. But Kube emphasised the program is not limited to these two properties alone.

propSym is open source and available for download for anyone with MATLAB. Darren Branch and Daniel Spencer, scientists from the Sandia National Labs, contributed to this research.

human brain so that it may be utilised to control a computer. Ildar Rakhmatulin and Sebastian Volkl, authors of an article on the arXiv open access server, advocate using a Raspberry Pi instead of hyper-expensive equipment for brain-computer interfacing.



Rakhmatulin studies BCIs at Russia's South Ural State University, and Volkl works in the disciplines of AI and neural networks. He previously created a laser turret for shooting down mosquitoes using a Raspberry Pi 3 B+ and camera. They banded together to form hackerBCI, a platform that attempts to make neuroscience projects and research more accessible.

PiEEG, a Raspberry Pi project, uses C, C++, and Python to read up to eight real-time electroencephalography (EEG) signals collected from the brain by electrodes put in the subject's cap. The pair has shared use cases on GitHub, including controlling robots and drones. One of the potential applications is the control of exoskeletons.

If you're interested in using your Raspberry Pi to read your brain signals, hackerBCI will shortly begin crowdfunding the PiEEG.

LibreOffice to include two 'made up' languages to 7.3 update

Starting in February this year, LibreOffice 7.3, an open source writing platform, will offer support for two 'made up' languages: Interslavic and Klingon.

According to *neowin.com*, the decision to incorporate the Klingon and Interslavic languages was made in order to reduce user workload by allowing them to work with the languages without having to use alternative translation. Linguist Marc Orkrand created the Klingon language for the Star Trek franchise. The goal of Interslavic, on the other hand, is to "bridge the language gap between Slavic languages such as Russian and Polish."

Eike Rathke, a Red Hat employee, added the language update to the newest edition of the LibreOffice software. The program can be tweaked to incorporate feedback from a variety of users' perspectives, resulting in a product that is essentially tuned to the work at hand by those who use it.

LibreOffice 7.3 will be released in February 2022. Visit *LibreOffice.org* for further information.

CAST releases new SCA browser extension for Chrome

CAST Highlight subscribers can now install and use a new browser extension that allows them to see the legal and security exposures associated with a particular open source component as they attempt to download it from the Internet.

The ubiquitous use of open source components in custom-built applications creates intellectual property and security risks for business owners and